

A S Prajwal Hiremath

Data Scientist | GenAI & ML Specialist

prajwalhiremath2017@gmail.com +91-9032847559 Bengaluru, Karnataka, India

[in linkedin.com/in/prajwal-hiremath-dev/](https://www.linkedin.com/in/prajwal-hiremath-dev/)

github.com/prajwalhiremath-dev

<https://prajwalhiremath.in/>

Professional Summary

Results-driven ML Engineer with 4+ years of experience architecting and deploying production AI systems. Currently at Deloitte USI, consulting on Anthropic Claude projects for pharmaceutical clients (Eli Lilly, Pfizer), delivering RAG systems and SQL agents with 95-99% accuracy. Previously at Cisco, architected enterprise platforms serving 300+ users. Strong full-stack capabilities combined with hands-on deep learning expertise across NLP, and audio processing.

Technical Skills

Programming and Languages : Python (Expert), SQL, Scikit-learn, Pandas, NumPy

Agentic AI & Frameworks : LangGraph, CrewAI, Langchain, FastMCP

GenAI & LLMs: RAG Architectures, Anthropic (Claude), OpenAI (GPT-4), Prompt Engineering

Data & Vector Databases: OpenSearch, FAISS, ChromaDB, PostgreSQL, Redis

Web & Infrastructure : FastAPI, Django, Flask, AWS (Bedrock, S3, SageMaker, RDS), Celery, Docker, Git

Deep Learning & ML : Transformers, PyTorch, CNNs, Audio ML, XGBoost, Model Evaluation, Feature Engineering

Professional Experience

Data Scientist (Consultant) – Deloitte USI

01/2025 – Present | Bengaluru, India

Client: Eli Lilly

SQL Agent with Langchain

- Architected an advanced SQL agent using Langchain and Anthropic Claude on AWS Bedrock to enable natural language querying of a 7-table, auto-refreshing AWS RDS (PostgreSQL) database.
- Engineered a dynamic few-shot learning system (using OpenSearch vector DB) and a specific Chain of Thought (CoT) prompt (intent-tables-joins-sql-validation) to boost domain accuracy by 18% and reduce hallucinations by 31%.
- Impact :** Achieved 95% query accuracy, outperforming OpenAI models and securing the client's continued engagement with Anthropic's technology.

Multi-Agent RAG with CrewAI

- Developed a 4-agent RAG pipeline (Query Router, Retriever, Synthesizer, Evaluator) using CrewAI to process complex natural language queries against clinical trial documents and SQL databases.
- Engineered a custom hybrid search engine (FAISS vector similarity + BM25 keyword matching) for high-precision retrieval and used DeepEval for automated relevancy evaluation, significantly improving research efficiency.
- Impact:** Significantly improved research efficiency by enabling natural language querying of complex scientific data.

Technologies: LangChain, CrewAI, Anthropic Claude, OpenAI, OpenSearch, FAISS, DeepEval, HuggingFace Embeddings, AWS BedRock, AWS Sagemaker

Client: Pfizer

OmniScout Multi-Tenant AI Platform

- Architected 'OmniScout,' a multi-tenant AI platform that consolidated four specialized agentic applications (OncoScout, VRDScout, etc.) into a single, scalable codebase, unifying RAG and SQL agent logic.
- Engineered a dynamic, header-driven configuration system using FastAPI Dependency Injection and Pydantic settings to load client-specific secrets and data sources based on the x-uc-client-id header.
- Developed client-level API rate-limiting middleware to ensure API stability and fair resource allocation across all use cases.
- Engineered a conversational AI tool using FastMCP with self-correcting validation and debugging agents, achieving 99% accuracy in translating natural language into complex SQL queries.
- Impact:** Drastically reduced code redundancy and maintenance overhead, unifying development and enabling the rapid deployment of new, specialized 'Scout' applications from a single source.
- Technology:** Python, FastAPI, Pydantic, LangGraph, AWS , Docker, PostgreSQL, FastMCP

Testbed Reservation System

- Architected a high-availability Django platform scaled to serve 300+ engineers and 5,000+ monthly reservations, eliminating 100% of double-booking race conditions by implementing PostgreSQL row-level locking within atomic transactions.
- Engineered a non-blocking Celery-Redis asynchronous notification system and built utilization dashboards, which together reduced testbed idle time by 45% (recovering \$150K in annual value).

ML Testbed Predictor

- Engineered a ML pipeline to predict testbed suitability, systematically training and benchmarking few classifiers including XGBoost, RandomForest.
- Implemented a robust preprocessing workflow using StandardScaler for feature scaling and LabelEncoder for categorical encoding, serializing the pipeline with joblib to prevent training-serving skew.
- Developed a Flask REST API to serve real-time predictions, leveraging comprehensive metrics (Accuracy, AUC-ROC, Precision-Recall) and visualizations (Confusion Matrix, ROC Curve) to validate model performance.

Cisco Wiki Analytics Dashboard

- Engineered a Flask and Pandas dashboard to automate executive reporting by ingesting, cleaning, and programmatically standardizing raw Excel data, which reduced manual report generation time by 40%.
- Delivered hierarchical, interactive chart views tailored for leadership, enabling Directors to see high-level aggregates and Managers to drill-down into specific team/engineer performance for data-driven decision-making.

Testbed Chatbot:

- Built a RAG chatbot (FastAPI, OpenAI Embeddings, ChromaDB) that enabled natural language queries for hardware specs, reducing information lookup time by 75%.

Technologies used : Django, PostgreSQL, Celery, Redis, Webex API, Flask, XGBoost, Scikit learn, Label Encoder, joblib, Pandas, FastAPI, OpenAI Embeddings, ChromaDB

Education

Bachelor of Technology in Electrical and Electronics Engineering | 8.11 CGPA
Jawaharlal Nehru Technological University, Anantapur

Personal Project

- Architected and trained five end-to-end PyTorch classification models across multiple domains (Audio, Image, Text, Tabular) with robust data pipelines, stratified validation, and standalone inference functions.
- Built custom deep learning architectures including: a 2D-CNN for audio classification with on-the-fly Mel Spectrogram generation; a multi-layer CNN with BatchNorm/Dropout for images; and a binary classification DNN for tabular data with sklearn preprocessing pipelines.
- Implemented transfer learning and NLP solutions: fine-tuned ResNet-18 by freezing backbone layers and retraining the classifier head; developed a sarcasm detector using custom classification layers on frozen BERT [CLS] embeddings with batch tokenization via collate_fn.

Achievements

- Secured \$2M+ pharmaceutical client engagement with 13% accuracy improvement over competitor LLMs
- Led testbed reservation system serving 300+ users, delivering \$650K+ cost savings with 99%+ uptime
- Built 5 production deep learning models across tabular, vision, audio, and NLP domains
- Reduced operational time by 40-80% through ML-powered automation and data-driven insights

Certifications

- CrewAI Multi Agent System – DeepLearning.AI
- Google IT Automation with Python – Coursera
- Introduction to Generative AI and LLMs – Google
- Python for Data Science and ML Bootcamp – Udemy
- The Complete SQL Bootcamp – Udemy

Declaration

I hereby declare that all the information provided in this resume is true and accurate to the best of my knowledge and belief.